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Mathematical methods in economics

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ANALYSIS OF 3D VEL GRAPHS WITH NEGATIVE VALUES OF VARIABLES WITH CALCULATION OF X4

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. $\text{Vel (GDPel)} = f(X1, X2, X3, X4, X5)$. In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D Vel graph when the values of the variables were as follows $X1 = X2 = X3 = 1$, $X4 = 0.76$, $X5 = -1$. In this example, Vel has constant values of 3.18. In the following Figure 2 the depicted 3D Vel graph increases 264.58 times at variables $X1 = X2 = X3 = 1$, $X4 = 0.29 \dots 0.76$, $X5 = -0.1 \dots -1$.

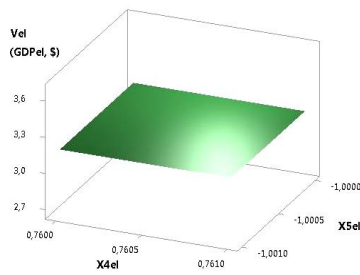


Fig. 1. 3D of Vel (GDPel)

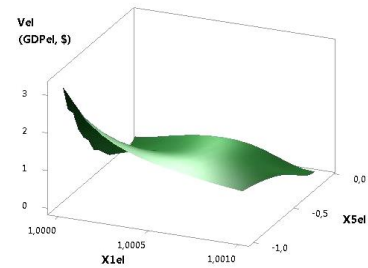


Fig. 2. 3D of Vel (GDPel)

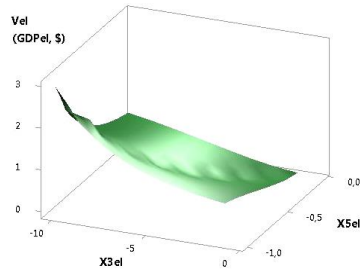


Fig. 3. 3D of Vel (GDPel)

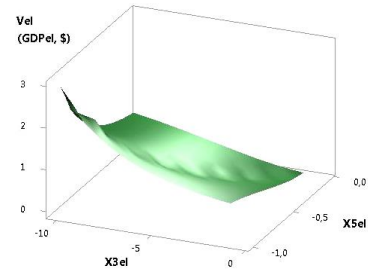


Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were $X1=1$, $X3=-1...-10$, $X4=0.14...0.7$, $X5=-0.1...-1$ and $X1=1$, $X2=X3=-1...-10$, $X4=0$, $X5=-0.1...-1$. As you can see, the plotted 3D Vel graph in Fig. 3 increases by 56.31 times, but in Fig. 4 it does not solution.

The calculated values for the 3D Vel graph in Figure 5 for variables $X1=X2=X3=-1...-10$, $X4=0.29...2.87$, $X5=-0.1...-1$ increases significantly by 1000 times. From the following Figure 6, it can be seen that for variables $X1=-1...-10$, $X2=X3=1$, $X4=0.65...0.19$, $X5=-1$, the Vel values decrease by 3.47 times.

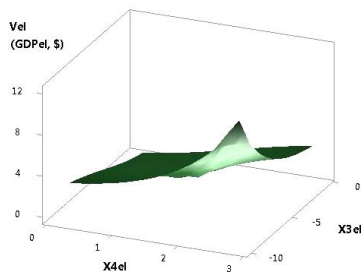


Fig. 5. 3D of Vel (GDPel)

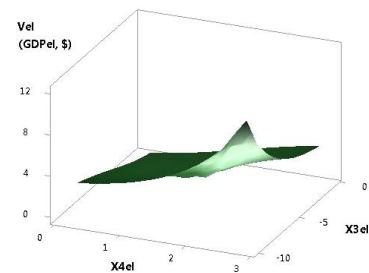


Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at $X1=X2=-1...-10$, $X3=1$, $X4=0.65$, $X5=-1$ and $X1=X2=X3=-1...-10$, $X4=0.76...2.87$, $X5=-1$. Here in Figure 7, parameter Vel has only one solution, and in Figure 8 it increases by 3.78 times.

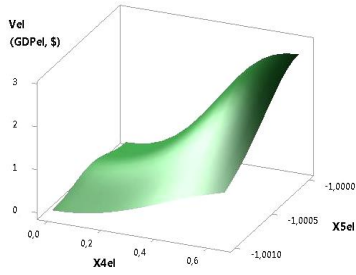


Fig. 7. 3D of Vel (GDPel)

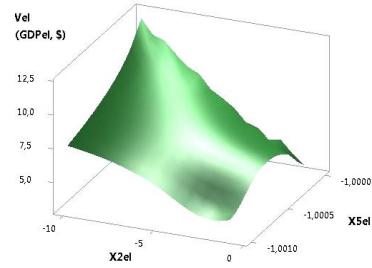


Fig. 8. 3D of Vel (GDPel)

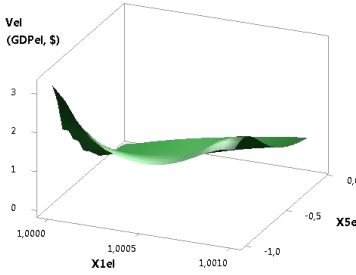


Fig. 9. 3D of Vel (GDPel)

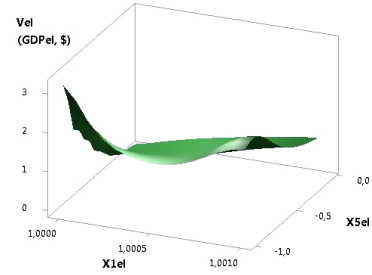


Fig. 10. 3D of Vel (GDPel)

The following figures 9 and 10 were plotted at $X_1 = X_2 = X_3 = 1$, $X_4 = 0.76 \dots 0.29$, $X_5 = -1 \dots -0.1$ and $X_1 = X_2 = 1$, $X_3 = -1 \dots -10$, $X_4 = 0.65 \dots 0.11$, $X_5 = -1 \dots -0.1$. Here in Figure 9, the parameter Vel decreases by 264.58 times, and in Figure 10 it decreases by 152.27 times.

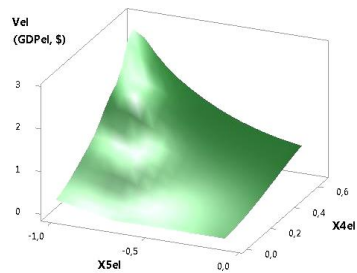


Fig. 11. 3D of Vel (GDPel)

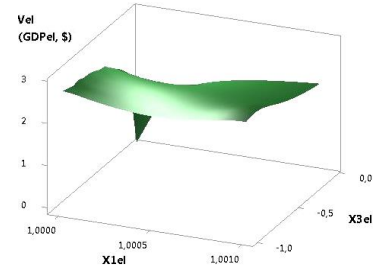


Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at $X_1 = 1$, $X_2 = X_3 = -1 \dots -10$, $X_4 = 0.65 \dots 0.3$, $X_5 = -1 \dots -0.1$ and $X_1 = X_2 = 1$, $X_3 = -1 \dots -0.1$, $X_4 = 0.65 \dots 0.34$, $X_5 = -1$. In these examples, in Figure 11, the parameter Vel is reduced by 3.41 times, and in Figure 12 it is reduced by 1.94 times.

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2. Negative mortgage rates. Now it's real. // <http://www.vestifinance.ru / articles/52690>
3. The mechanism of negative interest rates on loans // <http://kpk-bonus.ru/mexanizm-otricatelnyx-stavok-po-kreditam>

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PLOTTING OF 3D VEL GRAPHS WITH NEGATIVE VALUES OF VARIABLES BASED ON THE CALCULATION OF X4

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. $\text{Vel (GDPel)} = f(X1, X2, X3, X4, X5)$. In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D Vel graph when the values of the variables were as follows $X1= 1$, $X2= X3= -1 \dots -0.1$, $X4= 0.65 \dots 0.7$, $X5= -1$. In this case, the Vel increases by 1.08 times. In the following Figure 2 the depicted 3D Vel graph at variables $X1= X2= X3= -1 \dots -0.1$, $X4= 0.76 \dots 0.71$, $X5= -1$ decreases slightly by 1.07 times.

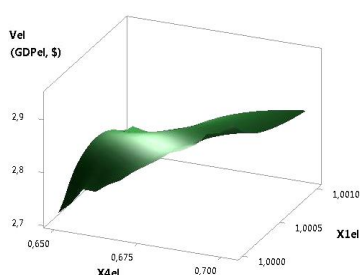


Fig. 1. 3D of Vel (GDPel)

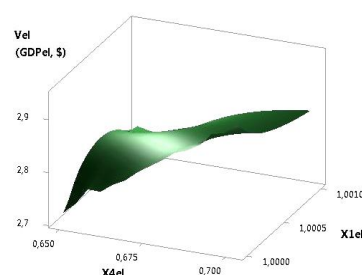


Fig. 2. 3D of Vel (GDPel)

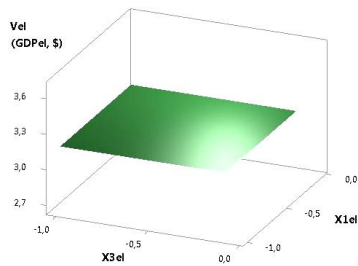


Fig. 3. 3D of Vel (GDPel)

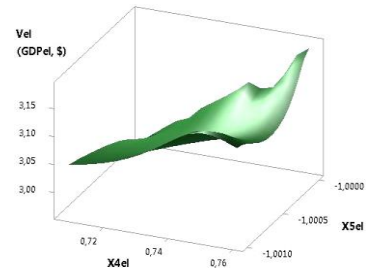


Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were $X1 = X3 = -1 \dots -0.1$, $X2 = 1$, $X4 = 0.76$, $X5 = -1$ and $X1 = X3 = 1$, $X2 = -1 \dots -0.1$, $X4 = 0.76 \dots 0.71$, $X5 = -1$. As you can see, the plotted 3D Vel graph in Fig. 3 has constant values of 3.18, and in Fig. 4 it decreases slightly by 1.07 times.

The calculated values for the 3D Vel graph in Figure 5 at variables $X1 = -1 \dots -0.1$, $X2 = X3 = 1$, $X4 = 0.65 \dots 0.7$, $X5 = -1$ increase slightly by 1.08 times. From the following Figure 6, it can be seen that at variables $X1 = X3 = 1$, $X2 = -1 \dots -10$, $X4 = 0.76 \dots 2.87$, $X5 = -1$, Val values increase by 3.78 times.

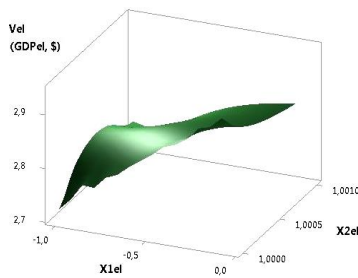


Fig. 5. 3D of Vel (GDPel)

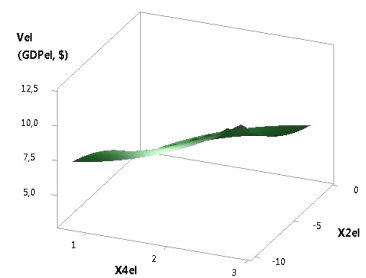


Fig. 6. 3D of Vel (GDPel)

The following figures 7 and 8 were plotted at $X1 = X2 = -1 \dots -10$, $X3 = 1$, $X4 = 0.76$, $X5 = -1$ and $X1 = X2 = 1$, $X3 = -0.1 \dots -1$, $X4 = 0.34 \dots 0.65$, $X5 = -1$. Here in Figure 7, the Vel values have constant values of 3.18, and in Figure 8 it increases by 1.94 times.

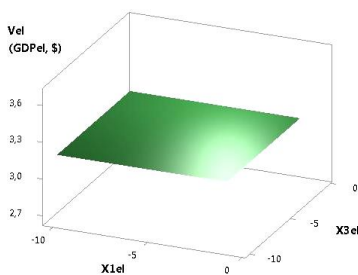


Fig. 7. 3D of Vel (GDPel)

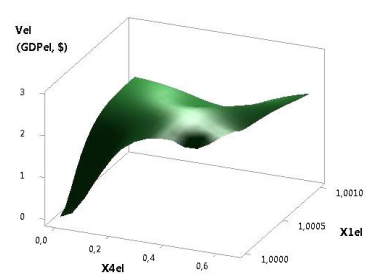


Fig. 8. 3D of Vel (GDPel)

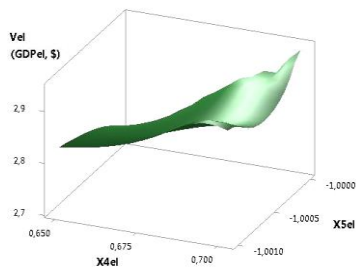


Fig. 9. 3D of Vel (GDPel)

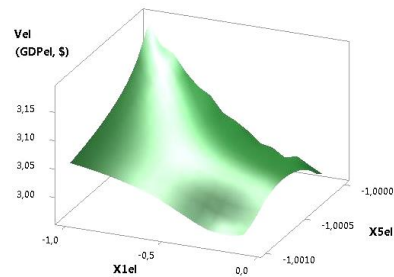


Fig. 10. 3D of Vel (GDPel)

The following Figure 9 was plotted at $X1=1$, $X2=X3=-0.1...-1$, $X4=0.7...0.65$ the Vel parameter decreases by 1.08 times. In Fig. 10, the plot at $X1=X2=X3=-0.1...-1$, $X4=0.71...0.76$, $X5=-1$ increases by 1.07 times.

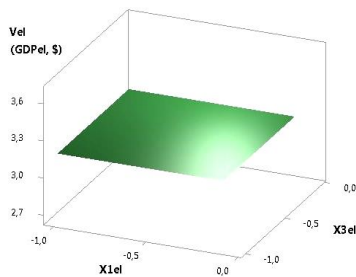


Fig. 11. 3D of Vel (GDPel)

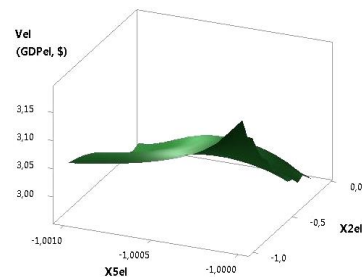


Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at $X1=X3=-0.1...-1$, $X2=1$, $X4=0.76$, $X5=-1$ and $X1=X3=1$, $X2=1$, $X4=0.71...0.76$, $X5=-1$. In these examples in Figure 11, the Vel variable remains constant 3.18, while in Figure 12 it increases slightly by 1.07 times.

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2. Negative mortgage rates. Now it's real. // <http://www.vestifinance.ru/articles/52690>
3. The mechanism of negative interest rates on loans // <http://kpk-bonus.ru/mexanizm-otricatelnyx-stavok-po-kreditam>

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THE EFFECT OF X4 CALCULATION ON THE PLOTTING OF 3D VEL GRAPHS WITH NEGATIVE VARIABLE VALUES

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. $Vel (GDP_{el}) = f(X1, X2, X3, X4, X5)$. In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D Vel graph when the values of the variables were as follows $X1 = -0.1 \dots -1$, $X2 = X3 = 1$, $X4 = 0.7 \dots 0.65$, $X5 = -1$. In this example, the Vel parameter is reduced by a factor of 1.08. In the following Figure 2 the depicted 3D Vel graph at variables $X1 = -1 \dots -10$, $X2 = 1$, $X3 = -0.1 \dots -1$, $X4 = 1.13$, $X5 = -1$ remains constant 4.73.

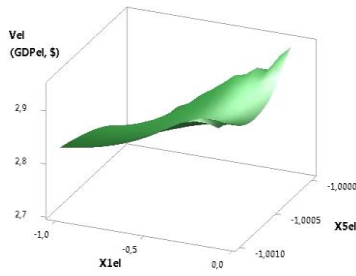


Fig. 1. 3D of Vel (GDPel)

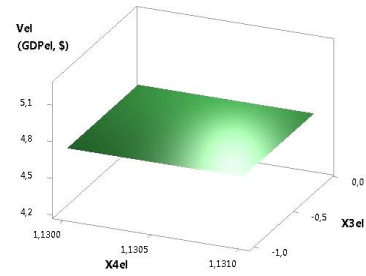


Fig. 2. 3D of Vel (GDPel)

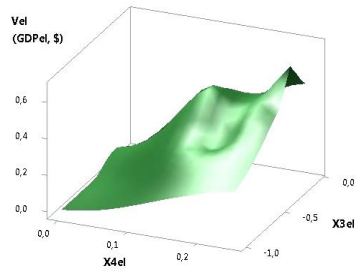


Fig. 3. 3D of Vel (GDPel)

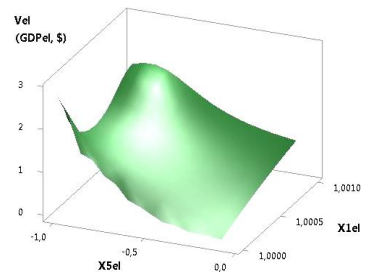


Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were $X_1 = X_2 = 1$, $X_3 = -0.1 \dots -1$, $X_4 = 0.13 \dots 0.16$, $X_5 = -1 \dots -0.1$ and $X_1 = 1$, $X_2 = X_3 = X_5 = -1 \dots -0.1$, $X_4 = 0.65 \dots 0.07$. As you can see, the plotted 3D Vel graph in Fig. 3 has a maximum of 0.67 at point 3, and in Fig. 4 it decreases by 242.24 times.

The calculated values for the 3D Vel graph in Figure 5 at variables $X_1 = X_2 = X_3 = X_5 = -1 \dots -0.1$, $X_4 = 0.76 \dots 0.08$ decrease significantly by 1000 times. From the following Figure 6, it can be seen that at variables $X_1 = X_3 = -0.1 \dots -1$, $X_2 = -0.1 \dots -1$, $X_4 = 0.76 \dots 0.29$, $X_5 = -1 \dots -0.1$, the 3D Vel graph decreases by 264.58 times.

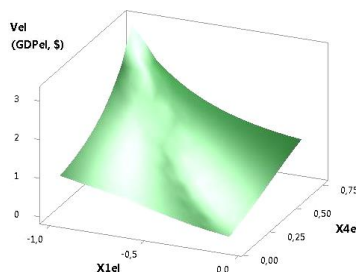


Fig. 5. 3D of Vel (GDPel)

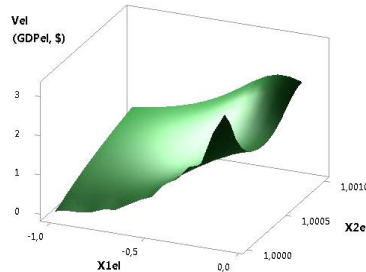


Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at $X_1 = X_3 = 1$, $X_2 = -0.1 \dots -1$, $X_4 = 0.71 \dots 0.29$, $X_5 = -1 \dots -0.1$ and $X_1 = X_5 = -1 \dots -0.1$, $X_2 = X_3 = 1$, $X_4 = 0.65 \dots 0.07$. Here in Figure 7, the Vel values decrease by 246.37 times, and in Figure 8 they decrease by 18.35 times.

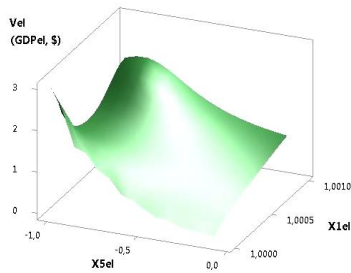


Fig. 7. 3D of Vel (GDPel)

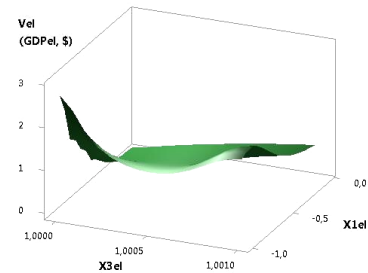


Fig. 8. 3D of Vel (GDPel)

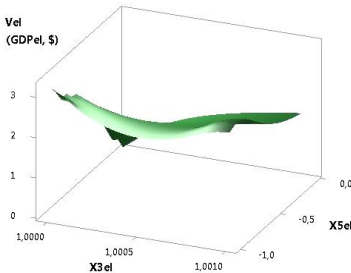


Fig. 9. 3D of Vel (GDPel)

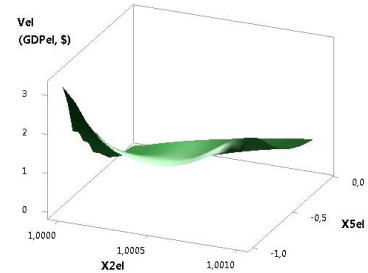


Fig. 10. 3D of Vel (GDPel)

The following Figure 9 was plotted at $X_1 = X_2 = 1$, $X_3 = -1 \dots -0.1$, $X_4 = 0.16 \dots 0.13$, $X_5 = -0.1 \dots -1$. Here, Vel decreases by 27.29 times. In Fig. 10, the Vel dependence plotted at $X_1 = X_3 = 1 \dots 10$, $X_2 = 1$, $X_4 = 0.76 \dots 0.29$, $X_5 = -1 \dots -0.1$ decreases by 264.58 times.

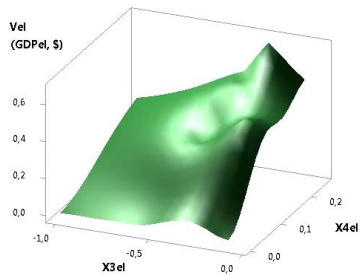


Fig. 11. 3D of Vel (GDPel)

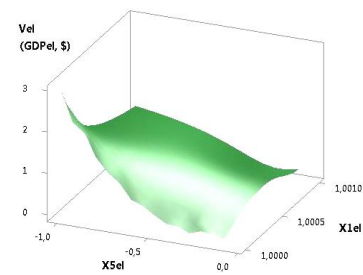


Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at $X_1 = 5 \dots -5$, $X_2 = X_3 = 1 \dots 10$, $X_4 = 0.99 \dots 0.1$, $X_5 = 10 \dots 1$ and $X_1 = X_5 = 5 \dots -5$, $X_2 = 1$, $X_3 = 10 \dots 1$, $X_4 = 0.99$. In these examples in Figure 11, the variable Val has a maximum of 0.67 at point 8, and in Figure 12 it increases by a factor of 27.33.

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2. Negative mortgage rates. Now it's real. // [http://www.vestifinance.ru / articles/ 52690](http://www.vestifinance.ru/articles/52690)
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Here, Figure 1 shows a 3D Vel graph when the values of the variables were as follows $X_1 = X_2 = X_3 = -1 \dots -0.1$, $X_4 = 0.29 \dots 0.71$, $X_5 = -0.1 \dots -1$. Here, the Vel parameter is increased by a factor of 246.37. In the following Figure 2 the depicted 3D Vel graph at variables $X_1 = X_3 = -1 \dots -0.1$, $X_2 = 1$, $X_4 = 0.29 \dots 0.76$, $X_5 = -0.1 \dots -1$, increases by 246.58 times.

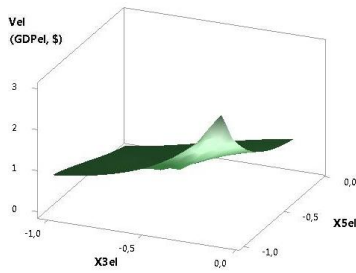


Fig. 1. 3D of Vel (GDPel)

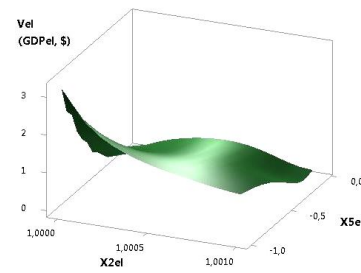


Fig. 2. 3D of Vel (GDPel)

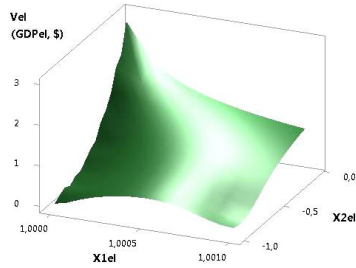


Fig. 3. 3D of Vel (GDPel)

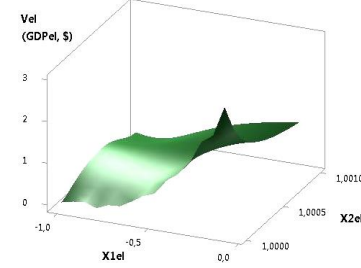


Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were $X1 = X3 = 1$, $X2 = -1 \dots -0.1$, $X4 = 0.29 \dots 0.71$, $X5 = -0.1 \dots -1$ and $X1 = -1 \dots -0.1$, $X2 = X3 = 1$, $X4 = 0.16 \dots 0.7$, $X5 = -0.1 \dots -1$. As you can see, the plotted 3D Vel graph in Fig. 3 increases by 246.37 times, and in Fig. 4 increases by 27.33 times.

The calculated values for the 3D Vel graph in Figure 5 at variables $X1 = X3 = 1$, $X2 = -1 \dots -10$, $X4 = 0.29 \dots 2.87$, $X5 = -0.1 \dots -1$ increase significantly by 1000 times. From the following Figure 6, it can be seen that at variables $X1 = X3 = -1 \dots -10$, $X2 = 1$, $X4 = 0.29 \dots 0.76$, $X5 = -0.1 \dots -1$, the parameter Vel increases by 264.58 times.

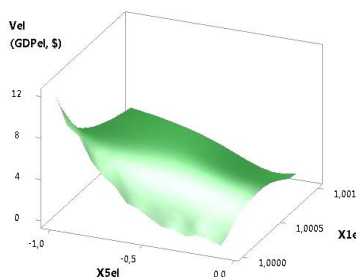


Fig. 5. 3D of Vel (GDPel)

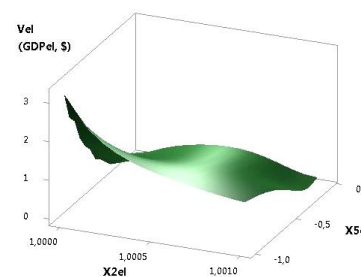


Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at $X1 = X2 = 1$, $X3 = -1 \dots -0.1$, $X4 = 0.65 \dots 7.02$, $X5 = 5 \dots -5$ and $X1 = X3 = 1 \dots 10$, $X2 = X5 = 5 \dots -5$, $X4 = -0.99 \dots -0.1$. Here in Figures 7 and 8 The Vel values increase by 1079.39 and 1078.77 times, respectively.

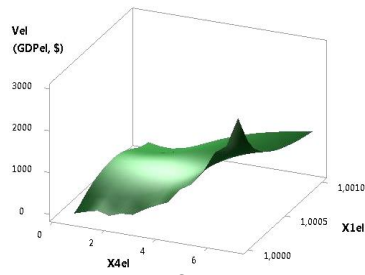


Fig. 7. 3D of Vel (GDPel)

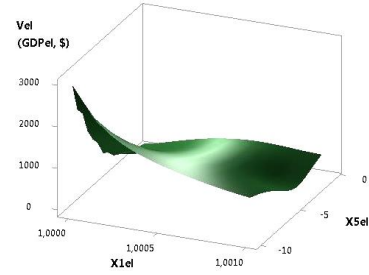


Fig. 8. 3D of Vel (GDPel)

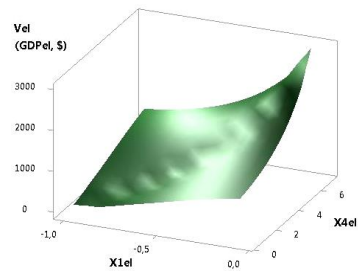


Fig. 9. 3D of Vel (GDPel)

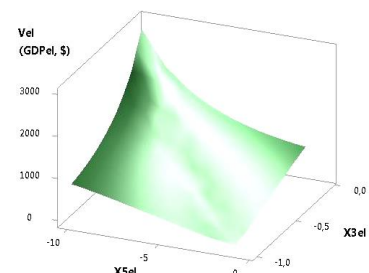


Fig. 10. 3D of Vel (GDPel)

The following Figure 9 was plotted at $X_1 = X_2 = X_3 = -1 \dots -0.1$, $X_4 = 0.76 \dots 7.07$, $X_5 = -1 \dots -10$ increases 930.49 times. In Fig. 10, the plot is plotted at and $X_1 = X_3 = -1 \dots -0.1$, $X_2 = 1$, $X_4 = 0.76 \dots 7.08$, $X_5 = -1 \dots -10$ increases by 931.21 times.

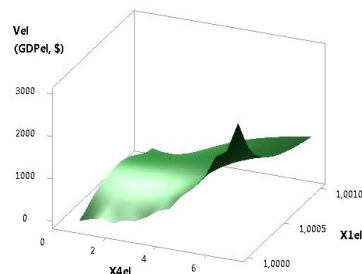


Fig. 11. 3D of Vel (GDPel)

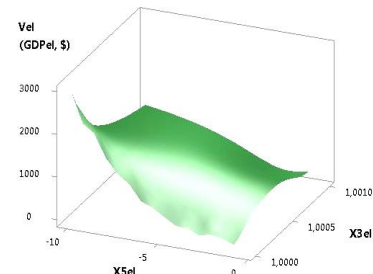


Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at $X_1 = X_3 = 1$, $X_2 = -1 \dots -0.1$, $X_4 = -0.76 \dots 7.07$, $X_5 = -1 \dots -10$, and $X_1 = -1 \dots -0.1$, $X_2 = X_3 = 1$, $X_4 = 0.65 \dots 7.07$, $X_5 = -1 \dots -10$. In these examples, in Figure 11, the parameter Vel increases by 930.49 times and in Figure 12 by 1087.77 times.

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ANALYSIS OF 3D VEL GRAPHS USING THE X4 VARIABLE AND WITH NEGATIVE VALUES OF OTHER VARIABLES

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. $Vel (GDPel) = f(X1, X2, X3, X4, X5)$. In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D graph of Vel, when the values of the variables were as follows $X1= X3= X5= -1...-10$, $X2= 1$, $X4= 0.76...7.08$. Here, the values of Vel increase by 931.21 times. In the following Figure 2 the depicted 3D Vel graph at variables $X1= X2= 1$, $X3= -1...-0.1$, $X4= 7.07...1.27$, $X5= -10...-1$ decrease by 139.1 times.

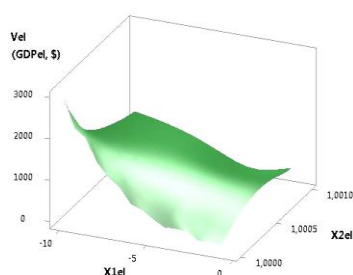


Fig. 1. 3D of Vel (GDPel)

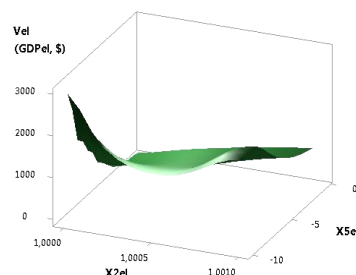


Fig. 2. 3D of Vel (GDPel)

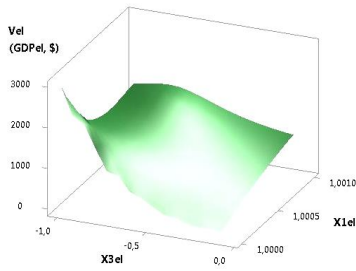


Fig. 3. 3D of Vel (GDPel)

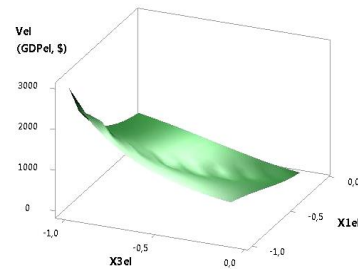


Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were $X_1=1$, $X_2= X_3=-1...-0.1$, $X_4= 7.07...0.7$, $X_5= -10...-1$ and $X_1= X_2=X_3= -1...-0.1$, $X_4= 7.07...0.71$, $X_5= -10...-1$. As we can see, the plotted 3D Vel graph in Fig. 3 decreases significantly by 1007.06 times, and in Fig. 4 by 1000 times.

The plotted 3D Vel graph in Figure 5 at variables $X_1= X_3=-1...-0.1$, $X_2= 1$, $X_4= 7.08...0.76$, $X_5= -10...-1$ decreases by 931.21 times. From the following Figure 6, it can be seen that at variables $X_1= X_3= 1$, $X_2= -1...-0.1$, $X_4= 7.08...0.71$, $X_5= -10...-1$, the Vel graph decreases 1000 times.

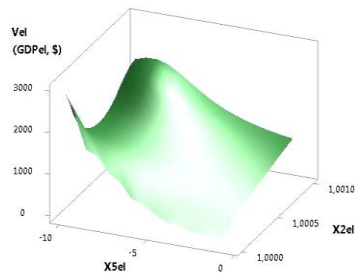


Fig. 5. 3D of Vel (GDPel)

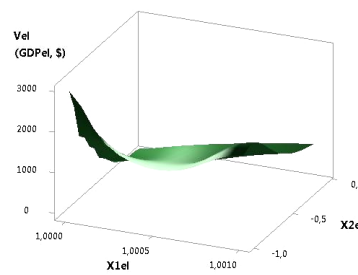


Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at $X_1=-1...-0.1$, $X_2= X_3= 1$, $X_4= 7.07...0.7$, $X_5= -10...-1$ and $X_1= X_3=-1...-10$, $X_2= 1$, $X_4=7.08...0.76$, $X_5= -10...-1$. Here in Figure 7, the Vel value decreases by a factor of 1007.06, and in Figure 8 it decreases by a factor of 931.21.

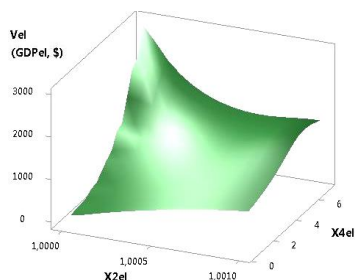


Fig. 7. 3D of Vel (GDPel)

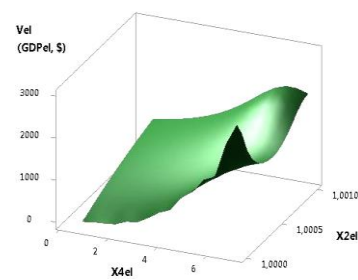


Fig. 8. 3D of Vel (GDPel)

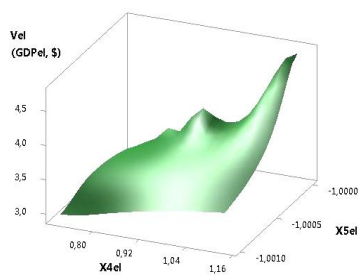


Fig. 9. 3D of Vel (GDPel)

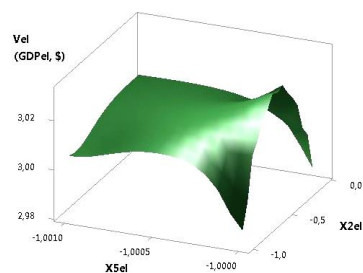


Fig. 10. 3D of Vel (GDPel)

The following Figure 9 was plotted at $X_1 = X_2 = -0.1 \dots -1$, $X_3 = -1 \dots -0.1$, $X_4 = 0.71 \dots 1.13$, $X_5 = -1$ increases slightly by 1.6 times. Figure 10 plotted at $X_1 = -0.1 \dots -1$, $X_2 = X_3 = -1 \dots -0.1$, $X_4 = 0.71 \dots 0.71$, $X_5 = -1$ has maxima of 3.03 at points 5 and 6.

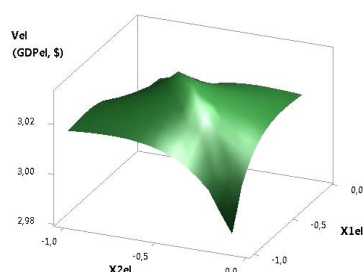


Fig. 11. 3D of Vel (GDPel)

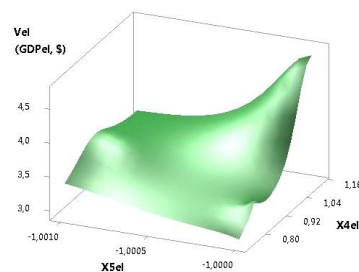


Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at $X_1 = -1 \dots -0.1$, $X_2 = X_3 = -0.1 \dots -1$, $X_4 = 0.71 \dots 0.71$, $X_5 = -1$ and $X_1 = X_2 = -1 \dots -0.1$, $X_3 = -0.1 \dots -1$, $X_4 = 1.13 \dots 0.71$, $X_5 = -1$. In these examples in Fig. 11, the Vel variable has maxima of 3.03 at points 5 and 6, and in Fig. 12 it decreases slightly by 1.6 times.

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3. The mechanism of negative interest rates on loans // <http://kpk-bonus.ru/mexanizm-otricatelnyx-stavok-po-kreditam>

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CALCULATION OF X4 AND PLOTTING OF 3D GRAPHS FOR VEL WITH NEGATIVE VALUES OF VARIABLES

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. $Vel (GDP_{el}) = f(X_1, X_2, X_3, X_4, X_5)$. In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D Vel graph when the values of the variables were as follows $X_1 = X_3 = -0.1 \dots -1$, $X_2 = -1 \dots -0.1$, $X_4 = 0.76 \dots 0.71$, $X_5 = -1$. Here Vel decreases by 1.07 times. In the following Figure 2 the depicted 3D Vel graph at variables $X_1 = X_3 = -1 \dots -0.1$, $X_2 = -0.1 \dots -1$, $X_4 = 0.71 \dots 0.76$, $X_5 = -1$ increases by 1.07 times.

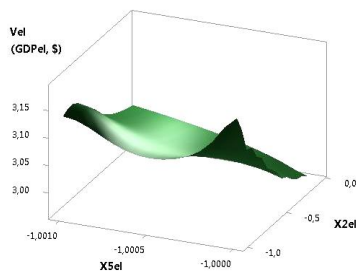


Fig. 1. 3D of Vel (GDPel)

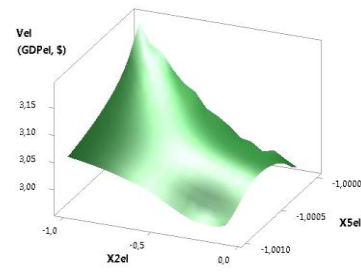


Fig. 2. 3D of Vel (GDPel)

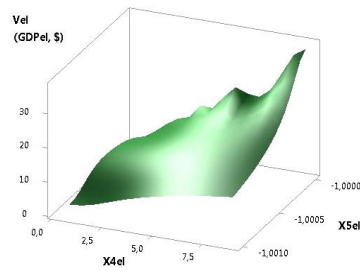


Fig. 3. 3D of Vel (GDPel)

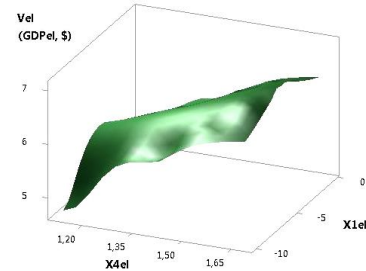


Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were $X1 = X2 = -1 \dots -10$, $X2 = -10 \dots -1$, $X4 = 0.71 \dots 8.83$, $X5 = -1$ and $X1 = X5 = 1 \dots 10$, $X2 = -1$, $X3 = 1 \dots 0.1$, $X4 = -0.1 \dots -0.99$. As you can see, the plotted 3D Vel graph in Fig. 3 increases by 12.39 times, and in Fig. 4 has maxima of 7.04 at points 5 and 6.

The plotted 3D Vel graph in Figure 5 at variables $X1 = -1 \dots -10$, $X2 = X3 = -1 \dots -10$, $X4 = 1.13$, $X5 = -1$ has maxima of 7.04 at points 5 and 6. From the following Figure 6 it can be seen that at variables $X1 = X2 = -10 \dots -1$, $X3 = -1 \dots -10$, $X4 = 8.83 \dots 0.71$, $X5 = -1$ Vel decreases by 12.39 times.

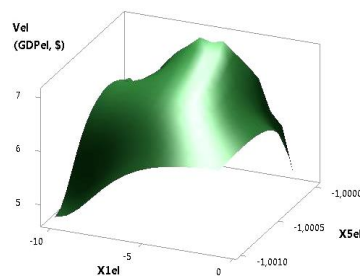


Fig. 5. 3D of Vel (GDPel)

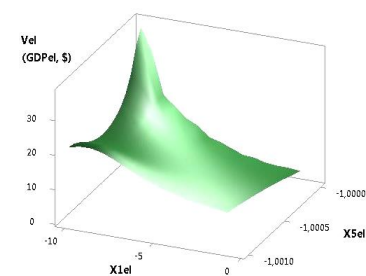


Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at $X1 = X3 = -1 \dots -10$, $X2 = -10 \dots -1$, $X4 = 2.87 \dots 0.76$, $X5 = -1$ and $X1 = X3 = -10 \dots -1$, $X2 = -1 \dots -10$, $X4 = 0.76 \dots 2.87$, $X5 = -1$. Here in Figure 7, the Vel values decrease by 3.78 times, and in Figure 8 they increase by 3.78 times.

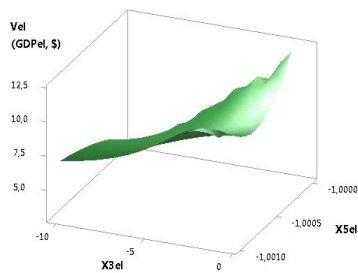


Fig. 7. 3D of Vel (GDPel)

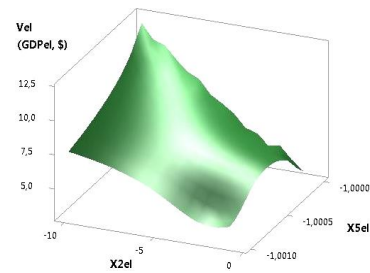


Fig. 8. 3D of Vel (GDPel)

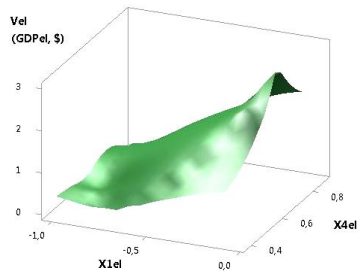


Fig. 9. 3D of Vel (GDPel)

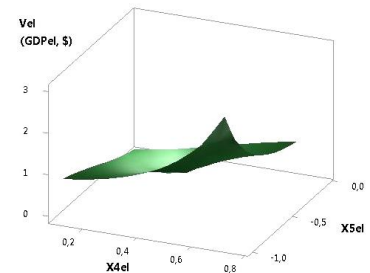


Fig. 10. 3D of Vel (GDPel)

The following Figure 9 was plotted at $X1 = X2 = -0.1 \dots -1$, $X3 = X5 = -1 \dots -0.1$, $X4 = 0.76 \dots 0.88$. Here the Vel parameter decreases by 80.07 times. The Vel parameter, built in Fig. 10 at $X1 = X2 = -1 \dots -10$, $X3 = 1 \dots 0.1$, $X4 = 0.99$, $X5 = 1 \dots 10$, decreases 631.06 times.

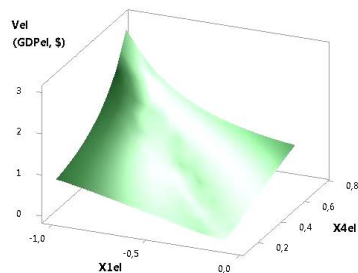


Fig. 11. 3D of Vel (GDPel)

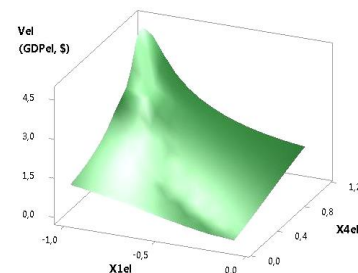


Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at $X1 = X5 = -1 \dots -0.1$, $X2 = X3 = -0.1 \dots -1$, $X4 = 0.71 \dots 0.11$ and $X1 = X2 = X5 = -1 \dots -0.1$, $X3 = -0.1 \dots -1$, $X4 = 1.13 \dots 0.07$. In these examples, in Figure 11, the Vel variable decreases by 631.06 times, and in Figure 12 it decreases by 942.52 times.

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3. The mechanism of negative interest rates on loans // <http://kpk-bonus.ru/mexanizm-otricatelnyx-stavok-po-kreditam>

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THE EFFECT OF THE VARIABLE X4 ON 3D GRAPHS FOR VEL WITH NEGATIVE VALUES OF VARIABLES

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. $\text{Vel (GDPel)} = f(X1, X2, X3, X4, X5)$. In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D Vel graph when the values of the variables were as follows $X1 = X3 = -0.1 \dots -1$, $X2 = X5 = -1 \dots -0.1$, $X4 = 0.76 \dots 0.1$. Here the Vel decreases significantly by a factor of 1000. In the following Figure 2 the depicted 3D Vel graph at variables $X1 = X3 = X5 = -1 \dots -0.1$, $X2 = -0.1 \dots -1$, $X4 = 0.71 \dots 0.29$ decreases by 246.37 times.

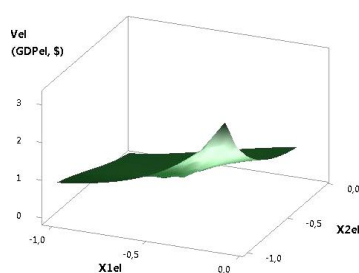


Fig. 1. 3D of Vel (GDPel)

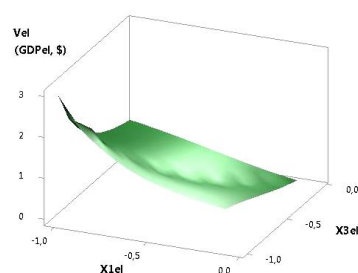


Fig. 2. 3D of Vel (GDPel)

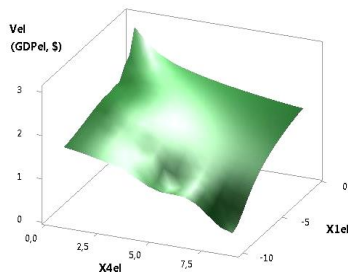


Fig. 3. 3D of Vel (GDPel)

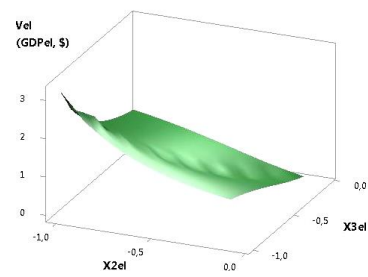


Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were $X1 = -10 \dots -1$, $X2 = -10 \dots -1$, $X3 = -10 \dots -1$, $X4 = 0.71 \dots 8.8$, $X5 = -1 \dots -0.1$ and $X1 = -1 \dots -10$, $X2 = -1 \dots -10$, $X3 = -1 \dots -10$, $X4 = 0.76 \dots 0.29$. As we can see, the plotted 3D Vel graph in Fig. 3 has a maximum of 2.02 at point 5, and in Fig. 4. it will decrease by 264.58 times.

The plotted 3D Vel graph in Figure 5 for variables $X1 = -10 \dots -1$, $X2 = -10 \dots -10$, $X4 = 1.13 \dots 0.88$, $X5 = -1 \dots -0.1$ decreases by 127.85 times. From the following Figure 6 it can be seen that for variables $X1 = -10 \dots -1$, $X2 = -10 \dots -1$, $X3 = -1 \dots -10$, $X4 = 0.883 \dots 0.11$, $X5 = -1 \dots -0.1$ Vel decreases very significantly by 7821.54 times.

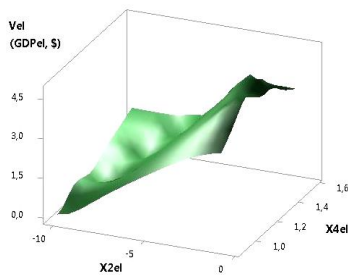


Fig. 5. 3D of Vel (GDPel)

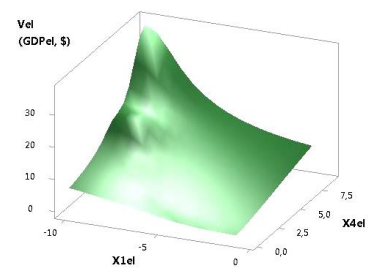


Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at $X1 = -10 \dots -1$, $X2 = -10 \dots -1$, $X4 = 2.87 \dots 0.29$, $X5 = -1 \dots -0.1$ and $X1 = -1 \dots -10$, $X2 = -1 \dots -10$, $X3 = -1$, $X4 = 1.13 \dots 0.29$, $X5 = -1 \dots -0.1$. Here in Figures 7 and 8, the Vel values decrease by 1000 times and 393.12 times, respectively.

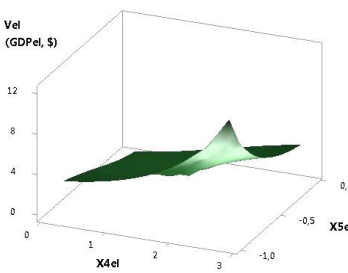


Fig. 7. 3D of Vel (GDPel)

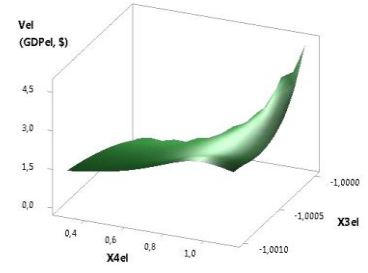


Fig. 8. 3D of Vel (GDPel)

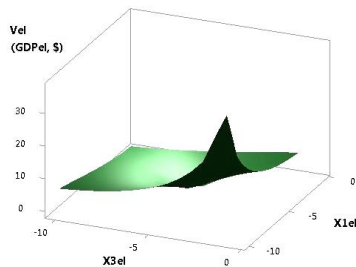


Fig. 9. 3D of Vel (GDPel)

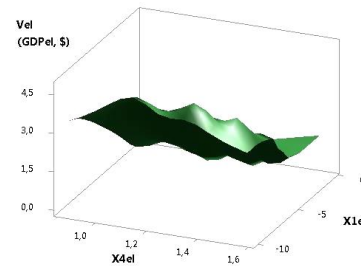


Fig. 10. 3D of Vel (GDPel)

The following Figure 9 was plotted at $X1 = X2 = -1 \dots -10$, $X3 = -10 \dots -1$, $X4 = 0.11 \dots 0.83$, $X5 = -0.1 \dots -1$, where Val increases significantly by 7821.54 times. Figure 10 plotted at $X1 = -1 \dots -10$, $X2 = X3 = -10 \dots -1$, $X4 = 0.8 \dots 1.13$, $X5 = -0.1 \dots -1$ shows that Val values increase by 127.85 times.

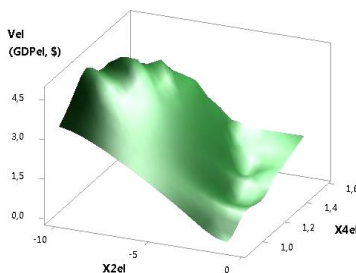


Fig. 11. 3D of Vel (GDPel)

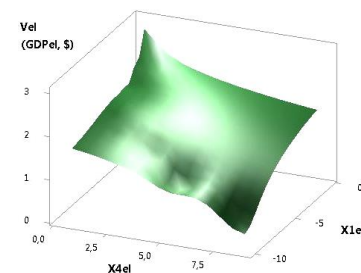


Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at $X1 = -10 \dots -1$, $X2 = X3 = -1 \dots -10$, $X4 = 0.88 \dots 1.13$, $X5 = -0.1 \dots -1$ and $X1 = X2 = -10 \dots -1$, $X3 = -1 \dots -10$, $X4 = 8.8 \dots 0.71$, $X5 = -0.1 \dots -1$. In these examples in Figure 11, the Vel variables increase by a factor of 127.85, and in Figure 12 they have a minimum of 2 at point 7.

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2. Negative mortgage rates. Now it's real. // <http://www.vestifinance.ru/articles/52690>
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HOW NEGATIVE VALUES OF VARIABLES AFFECT X4 CALCULATIONS AND THE PLOTTING OF 3D GRAPHS VEL

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. $Vel (GDP_{el}) = f(X_1, X_2, X_3, X_4, X_5)$. In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D Vel graph when the values of the variables were as follows $X_1 = X_3 = -1 \dots -10$, $X_2 = -10 \dots -1$, $X_4 = 2.78 \dots 0.76$, $X_5 = -0.1 \dots -1$. In this example, Vel is exaggerated by 27.29 times. In the following Figure 2 the depicted 3D Vel graph at variables $X_1 = X_3 = -10 \dots -1$, $X_2 = -1 \dots -10$, $X_4 = 0.29 \dots 2.87$, $X_5 = -0.1 \dots -1$ increases 1000 times.

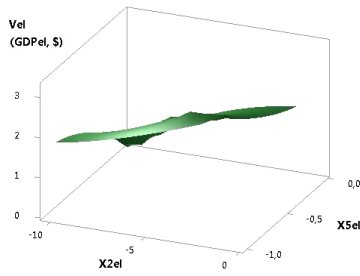


Fig. 1. 3D of Vel (GDPel)

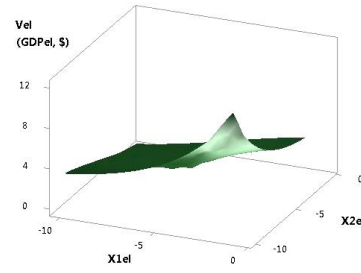


Fig. 2. 3D of Vel (GDPel)

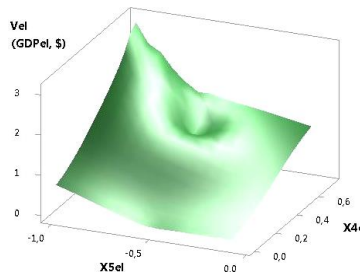


Fig. 3. 3D of Vel (GDPel)

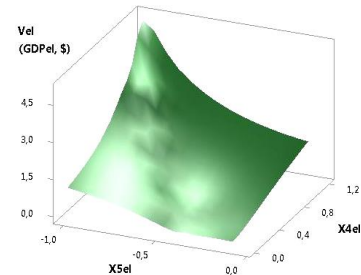


Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were $X1=5 \dots -5$, $X2=-10 \dots -1$, $X3=-1 \dots -10$, $X4=0.59 \dots 0.73$, $X5=-0.1 \dots -1$ and $X1=X2=5 \dots -5$, $X3=-1 \dots -10$, $X4=0.35 \dots 1.21$, $X5=-0.1 \dots -1$. As you can see, the plotted 3D Vel graph in Fig. 3 increases by 3.41 times, and in Fig. 4 increases by 13.84 times.

The calculated values for the 3D Vel graph in Figure 5 at variables $X1=X2=X3=5 \dots -5$, $X4=1.39 \dots 1.56$, $X5=-0.1 \dots -1$ increase by 111.84 times. From the following Figure 6, it can be seen that at variables $X1=X3=5 \dots -5$, $X2=1.10$, $X4=0.29 \dots 2.87$, $X5=-0.1 \dots -1$, Val values increase 1000 times.

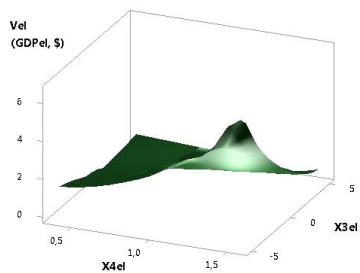


Fig. 5. 3D of Vel (GDPel)

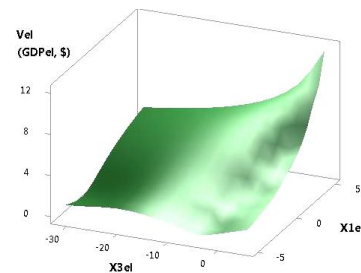


Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at $X1=1 \dots 10$, $X2=5 \dots -5$, $X3=-1 \dots -10$, $X4=-0.32 \dots 0.18$, $X5=-0.1 \dots -1$ and $X1=-10 \dots -1$, $X2=5 \dots -5$, $X3=-1 \dots -10$, $X4=4.4 \dots 0.83$, $X5=-0.1 \dots -1$. Here in Figure 7, the Vel values have a maximum of 0.6 at point 6, and in Figure 8 they increase by 18.91 times.

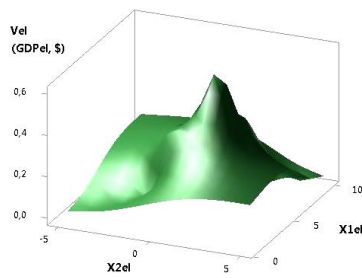


Fig. 7. 3D of Vel (GDPel)

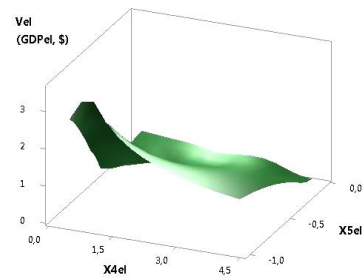


Fig. 8. 3D of Vel (GDPel)

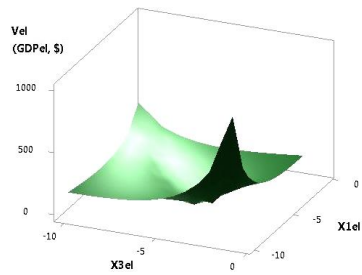


Fig. 9. 3D of Vel (GDPel)

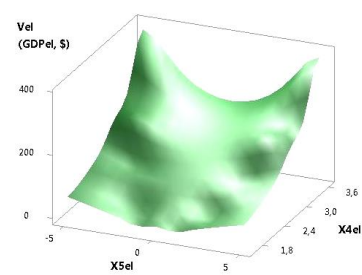


Fig. 10. 3D of Vel (GDPel)

The following figures 9 and 10 were plotted at $X_1 = X_2 = -10 \dots -1$, $X_3 = -1 \dots -10$, $X_4 = 9.49 \dots 3.53$, $X_5 = 5 \dots -5$ and $X_1 = -10 \dots -1$, $X_2 = X_3 = -1 \dots -10$, $X_4 = 3.64 \dots 3.64$, $X_5 = 5 \dots -5$. Here in Figure 9, the Vel variable has a minimum of 1.63 at point 6, and in Figure 10 it has a minimum of 1.99 at point 6.

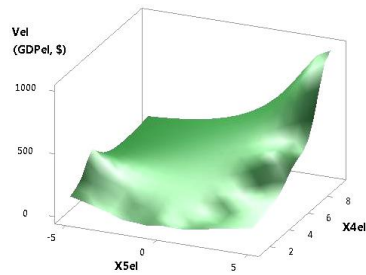


Fig. 11. 3D of Vel (GDPel)

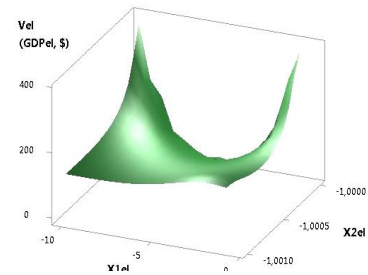


Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at $X_1 = X_2 = -1 \dots -10$, $X_3 = -10 \dots -1$, $X_4 = 3.54 \dots 9.48$, $X_5 = 5 \dots -5$ and $X_1 = -1 \dots -10$, $X_2 = -1$, $X_3 = -10 \dots -1$, $X_4 = 3.54 \dots 3.64$, $X_5 = 5 \dots -5$. In these examples in Figure 11, the Vel variable has a minimum of 1.75 at point 5, and in Figure 12 it has a minimum of 0.62 at point 5.